

Technical Document #934: Data Engineering - Comprehensive Overview

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Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Data lineage tracks the origin and movement of data through systems.
- Stream processing handles continuous data streams in real-time.
- Data governance establishes policies and procedures for managing data assets.